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Installing Stage for ROS2

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Installation of stage_ros2 on Ubuntu 24.04: (not tested in all configurations)

- Update the needed development packages and tools:

```
– sudo apt-get install git cmake g++ libjpeg8-dev \
    libpng-dev libglu1-mesa-dev libltdl-dev libfltk1.3-dev
```

– install FLTK of version 1.3 instead of 1.1 here

- Install stage as dependency for the stage_ros2 bridge: (only works manually on Ubuntu 24.04)

```
– cd YOUR_ROS2_WORKSPACE && mkdir src && cd src
  git clone --branch ros2 https://github.com/tuw-robotics/stage.git
  cd stage
  mkdir build && cd build
  cmake ..
  make # <- this will take a while
  sudo make install # <- will install the needed cmake files for us
```

– Use make -j<number_of_cores> to speed it up make (not recommended on wsl2)

- Make sure stage_ros2 finds the stage installation:

```
– export CMAKE_PREFIX_PATH=/usr/local/lib/cmake:$CMAKE_PREFIX_PATH
```

– this path might be different on your system or not needed when installing in the same workspace

- Install stage_ros2: Either as a submodule or as a normal git clone:

```
– cd YOUR_ROS2_WORKSPACE/src
  git clone --branch humble https://github.com/tuw-robotics/Stage.git
  cd YOUR_ROS2_WORKSPACE
```

– or as a submodule:

```
– cd <workspace>/src
  git submodule add --branch humble https://github.com/tuw-robotics/stage_ros2.git stage_ros2
```

- Make sure the python ros development tools are installed:

```
– cd YOUR_ROS2_WORKSPACE
  sudo apt install python3-rosdep
  sudo rosdep init
  rosdep update
  rosdep install --from-paths src --ignore-src -r -y
```

- Build the workspace:

```
– colcon build --symlink-install --cmake-args -DOpenGL_GL_PREFERENCE=LEGACY
  colcon build --symlink-install --packages-select stage_ros2
```

- Source Workspace and run the stage_ros2 example:

```
– source install/setup.bash
  ros2 launch stage_ros2 stage.launch.py world:=cave \
    enforce_prefixes:=false one_tf_tree:=true
```

Full example of the stage_ros2 example:

```
# Install dependencies
sudo apt-get install git cmake g++ libjpeg8-dev libpng-dev \
  libglu1-mesa-dev libltdl-dev libfltk1.3-dev

# Build and install stage
cd YOUR_ROS2_WORKSPACE
mkdir src
cd src
git clone --branch ros2 https://github.com/tuw-robotics/Stage.git
cd stage
cmake -S . -B build
cd build
make
sudo make install # install cmakefiles
export CMAKE_PREFIX_PATH=/usr/local/lib/cmake:$CMAKE_PREFIX_PATH # maybe not needed

# Install stage_ros2
cd YOUR_ROS2_WORKSPACE/src
git clone --branch humble https://github.com/tuw-robotics/stage_ros2.git
cd YOUR_ROS2_WORKSPACE

# Update the needed development packages and tools
sudo apt install python3-rosdep
sudo rosdep init
rosdep update
rosdep install --from-paths src --ignore-src -r -y

# Build the workspace
colcon build --symlink-install --cmake-args -DOpenGL_GL_PREFERENCE=LEGACY
colcon build --symlink-install --packages-select stage_ros2
```

- How to use it in other packages:
 - For easy access add sourcing of the workspace setup to the bash


```
* bash      echo "source ~/YOUR_ROS2_WORKSPACE/install/setup.bash" >> ~/.bashrc
```